

HAT-P-36b

R-1855

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_150327 · created 2026-08-03T08:44:26+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457108.88481 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1523 +/- 0.0039
Transit depth (Rp/Rs)^2	2.32 +/- 0.12 [%]
Orbital Inclination (inc)	86.92 +/- 1.43
Airmass coefficient 1 (a1)	1.0278 +/- 0.0072
Airmass coefficient 2 (a2)	-0.0234 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	#3 - [196, 332]
Optimal Aperture	3.44
Optimal Annulus	10.47
Transit Duration (day)	0.0977 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1523 $\pm$ 0.0039 vs 0.126 (deviation 6.24 σ)
Duration fit vs published	0.0977 d vs 0.0925 d (deviation 2.19 σ)
Mid-time O-C	-1.9 min
Depth SNR	36.1
Residual scatter	0.7 %

Light curve

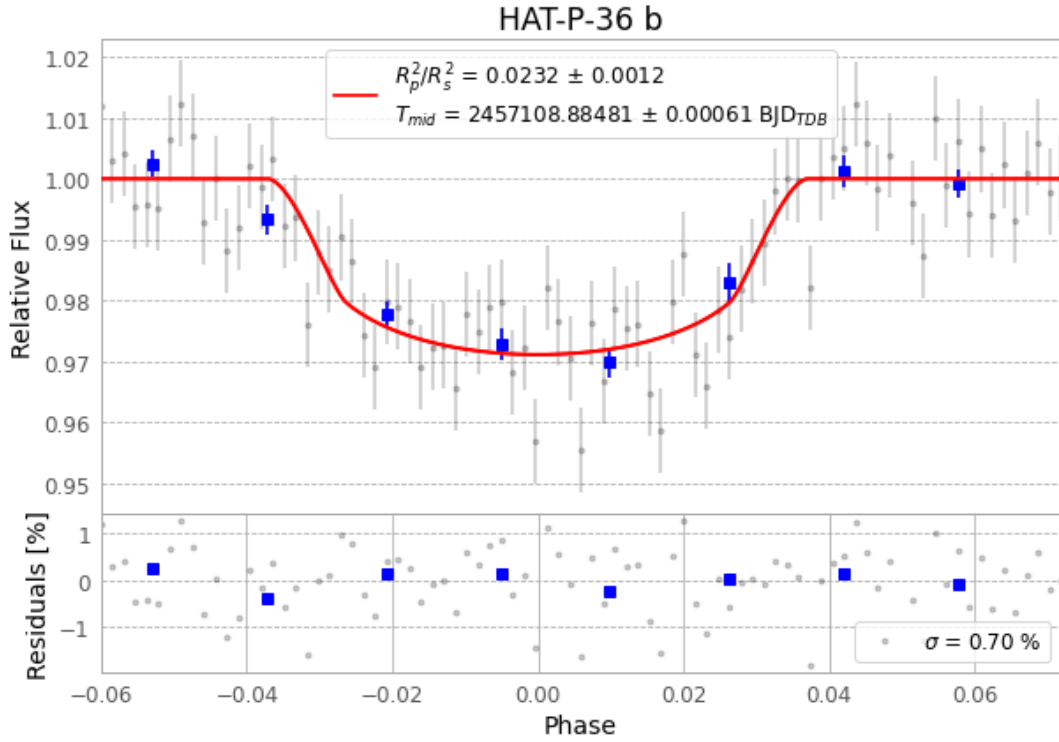


Figure 1 — FinalLightCurve_HAT-P-36 b_2015-03-27.png (SHA-256 e9676d96bb498a1d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a6deadcccd0a0a91...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-36150327071212.FITS → HATP-36150327112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-150327.log (4ffb7deae0f7...), FinalLightCurve_HAT-P-36 b_2015-03-27.png (e9676d96bb49...), FinalLightCurve_HAT-P-36 b_2015-03-27.csv (bd80ba202ad9...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:44Z.